Cross-Domain Morphological Classification

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Abstract

Accurate identification of immune cells within tissue and blood samples is critical for disease diagnosis, patient stratification, and prognosis assessment. While deep learning models have achieved high performance in automated cell classification within specific imaging domains, it remains unclear whether morphological features learned in one biological context can transfer to other modalities. In this project, we investigate whether a machine learning model trained on hematoxylin and eosin (H&E)-stained tissue derived morphologies can accurately distinguish immune from non-immune cells. Specifically, we evaluate cross-domain generalization without additional training to assess whether immune cell morphology contains domain-invariant features or depends on is highly context-dependent. Our results show that models trained on H&E images fail to generalize under cross-domain validation, indicating a substantial domain gap.

1 Introduction

Accurate identification of immune cell populations within biological samples is essential for disease diagnosis, patient stratification, and treatment planning. Numerous studies have demonstrated that the presence, abundance, and spatial distribution of immune cells within tissue are strongly associated with clinical outcomes [1,2,3]. For example, increased infiltration of CD8+ T cells has been correlated with improved prognosis in colorectal cancer [1], and the ratio of CD4+ to CD8+ tumor-infiltrating lymphocytes has shown prognostic value in breast cancer [3]. These findings underscore the importance of reliable identification of immune cells from surrounding non-immune cells in both research and clinical settings.

In routine pathology practice, this distinction is largely based on morphological assessment of stained microscopy images, such as hematoxylin and eosin (H&E)-stained tissue sections or stained peripheral blood smears (PBS). Pathologists rely on features including cell size, nuclear shape, chromatin pattern, and cytoplasmic properties to characterize cell identity. However, manual evaluation is time-consuming and subjective, particularly in complex tissue microenvironments where multiple cell types coexist. Recent advances in machine learning offer the potential to quantitatively capture morphological patterns from cells for the automation of cell classification with increased consistency and performance, steadily approaching expert-level accuracy.

Although both H&E and PBS are microscopy images, solid and liquid biopsies contain fundamental differences in the visual information they provide, such as tissue structure and cell morphology, even across a single cell type with recognizable features like immune cells. *Therefore, we hypothesize that immune cell classification relies on both intrinsic morphology and domain-specific contextual features, and that models trained on H&E tissue slides will fail to fully generalize to PBS microscopy images due to domain discrepancies.*

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2 Related Works

Foundation models in computational pathology are large-scale self-supervised models trained on millions of histology image patches to learn generalizable, morphological features. These learned features can then be fine-tuned for downstream tasks such as diagnosis, prognosis prediction, and cancer grading [4, 5, 6].

UNI [7] is a foundation model trained on hundreds of millions of H&E and immunohistochemistry (IHC) images, using a ViT-L-16 (large, 16-head) vision transformer (ViT) architecture [8] for powerful, generic feature extraction on whole-slide images. UNI, and its successor UNI2-h built upon a much larger ViT, outperform the state-of-the-art models in region-of-interest (ROI) identification and cell/tumour type classification tasks. DinoBloom [9] is another foundation model trained extensively on blood cell images to extract rich single-cell representations. Using self-supervised learning, it captures complex morphological features without requiring manual annotations, able to capture biologically meaningful information to improve the performance of other downstream bioinformatics models and general-purpose imaging models.

Neither models have been evaluated on other imaging modalities beyond its own training domain, and have only been benchmarked against other models within its competitive market. It remains unclear whether a hematology-specific foundation model can capture fundamental cellular morphology that generalizes across tissue contexts and staining modalities. Investigating the performance of UNI on liquid biopsies and DinoBloom on solid tumor H&E images can therefore help assess whether morphological representations of immune cells are shared across imaging and tissue domains.

3 Methods & Algorithm

Data Processing. We gathered a dataset for each imaging modality, ensuring that both were carefully annotated for a wide variety of cell types through orthogonal biological validation. The *Immunocto* database contains segmented H&E slides of 2 million immune and 4.5 million other (non-immune) cells, annotated using co-localized immunofluorescence (IF)-guided labeling and subsequent manual review [10]. The *Acute Promyelocytic Leukemia (APL)* dataset contains over 2,500 microscopy images of peripheral blood smear (PBS) cells spanning multiple hematopoietic lineages. In this dataset, cell types are grouped into two categories: immune cells (e.g., lymphocytes, monocytes) and non-immune cells (e.g., erythroblasts and other precursor cell types). Images from this collection were stained using the May–Grunwald–Giemsa method, a standard staining method used for hematological pathology. To enable cross-domain comparison, annotations in both dataset were consolidated into a binary classification of immune and non-immune cells. Due to its large scale, the *Immunocto* dataset is randomly shuffled and downsampled to 100,000 images (Table 1).

Table 1: Total number and proportion of cell types in both datasets

Count and proportion of cell types			
Dataset	Immune (%)	Non-Immune (%)	Total
Immunocto	$33,314$ (33%)	$66,686$ (67%)	$100,000$
APL	$8,976$ (94%)	584 (6%)	9560

To address the discrepancy in dataset sizes, we performed sampling with replacement and fixed the number of training samples per epoch. Furthermore, as both datasets exhibited substantial class imbalance, we employed the PyTorch algorithm `WeightedRandomSampler` to balance the distribution during training. Specifically, each sample is assigned a weight inversely proportional to its class frequency, allowing rarer classes to be sampled more often. We further included random rotation-based data augmentation to improve robustness and avoid overfitting. These approaches altogether allowed the model to have a balanced representation for learning and improved robustness under highly skewed data conditions.

Images in the *Immunocto* dataset are patched and centered around each individual cell, paired with a boolean mask outlining the morphology of the cell. This relieves a lot of the segmentation legwork typically required by automated cell-type classifiers in histopathological imaging. The *APL* dataset, unfortunately, did not provide cell masks for each sample. Cellpose-SAM, a deep learning–based

instance segmentation framework optimized for biological images [11], was used to generate the masks for each image prior to any training instances in order to avoid disparities between the domains. All images were re-sized and grey-scaled for normalization.

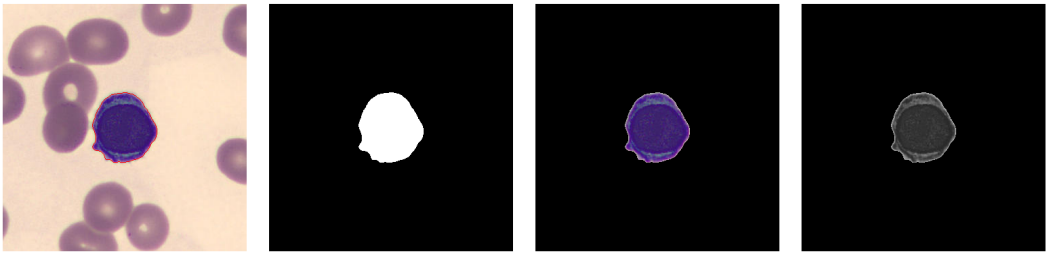

Figure 1: From left to right: lymphocyte blood smear in the *APL* dataset, computed Cellpose-SAM segmentation mask, masked image, and masked grey-scaled image for normalization.

Model Architecture. For our binary classification problem, we considered three models: (1) a ResNet18 residual neural network [12], (2) a ViT-L-16, and (3) an ensemble model leveraging the predictive powers of both. The first two architectures have found universal success in a variety of image classification tasks. Both ResNet and ViTs are gold-standards for image feature extraction still commonly-used in a variety of tasks, including immune cell sub-type stratification in the original *Immunocto* article and the basis for the UNI foundation model. As their approaches to retrieving information from images are fundamentally different, we wanted to further investigate whether one model provided unique strengths over the other through direct comparison and coupling in the ensemble model.

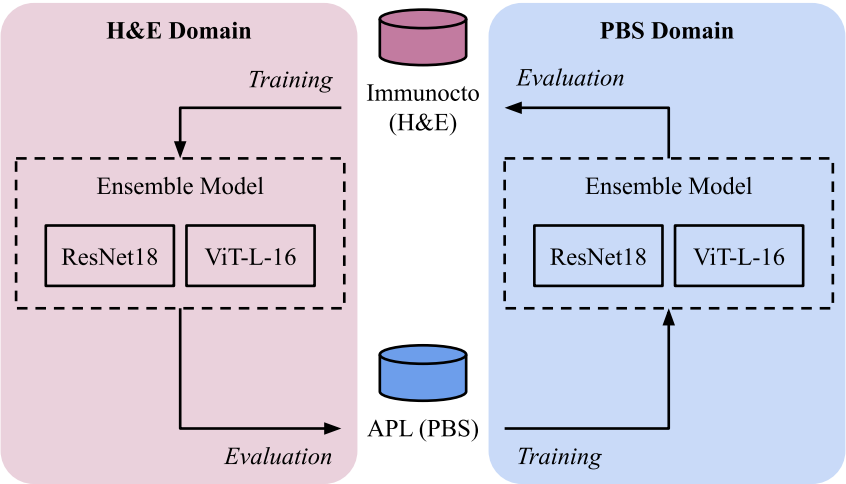

Figure 2: Workflow diagram of ensemble model training and cross-domain validation.

4 Results

Each of the 3 models were trained on both datasets independently, yielding a total of 6 fully-trained models. We then trained two baseline foundation models, UNI for H&E and DinoBloom for PBS, by freezing the weights for feature extraction and training a single fully-connected layer for classification. The training loss, training accuracy, and validation accuracy of all 6 models per epoch are plotted in Figure 4. The overall accuracy, precision, recall, and F1-score were computed on the testing set in both imaging domains to measure cross-domain performance. The baseline foundation models are italicized and best overall performing model is highlighted in bold, see Table 2.

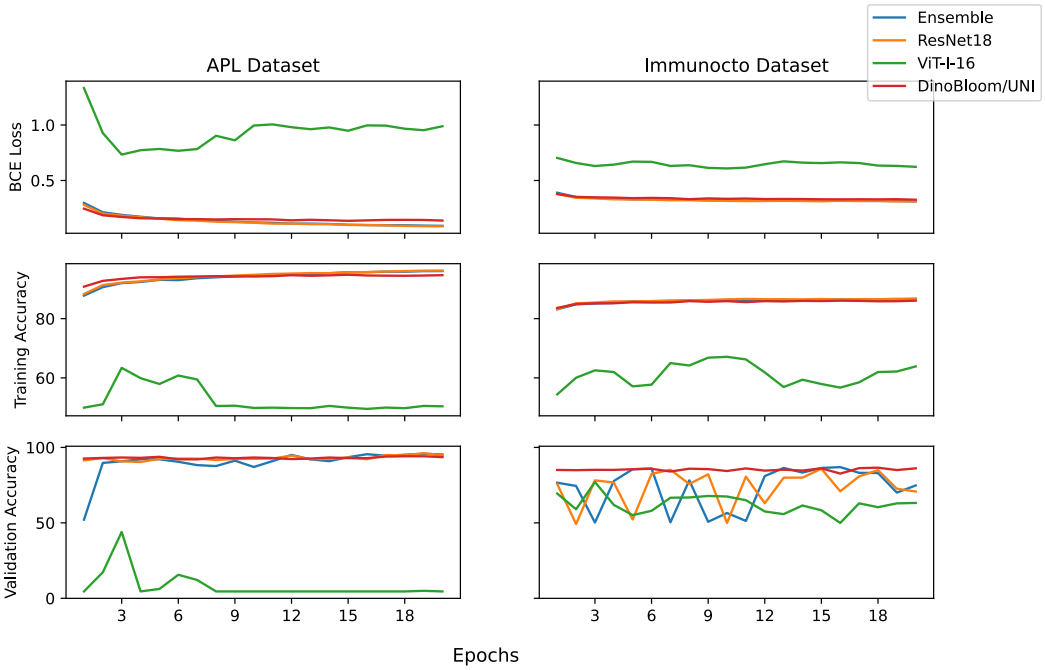

Figure 3: Loss and accuracy of models over training epochs.

Table 2: Comparison of performance between model architectures.

Intra-Domain Classification				
Name	Accuracy	Precision	Recall	F1 Score
APL-Resnet18	0.936	0.977	0.954	0.962
APL-ViT-L-16	0.056	0.000	0.000	0.000
APL-Ensemble	**0.947**	0.981	**0.963**	**0.972**
APL-DinoBloom	0.941	**0.986**	0.951	0.968
Immunocto-Resnet18	**1.000**	**1.000**	**1.000**	**1.000**
Immunocto-ViT-L-16	0.800	1.000	0.667	0.800
Immunocto-Ensemble	0.700	0.625	1.000	0.769
Immunocto-UNI	1.000	1.000	1.000	1.000

Cross-Domain Classification				
Name	Accuracy	Precision	Recall	F1 Score
APL-Resnet18	0.600	0.556	1.000	0.714
APL-ViT-L-16	0.400	0.000	0.000	0.000
APL-Ensemble	0.600	0.600	1.000	0.750
APL-DinoBloom	**0.800**	**0.778**	**1.000**	**0.875**
Immunocto-Resnet18	0.371	0.934	0.359	0.519
Immunocto-ViT-L-16	**0.484**	**0.951**	**0.478**	**0.636**
Immunocto-Ensemble	0.390	0.940	0.378	0.539
Immunocto-UNI	0.204	0.847	0.191	0.312

5 Discussion

Intra-Domain Classification. Models trained on PBS achieved high performance across all architectures, indicating that immune cell classification in blood smears is primarily driven by intrinsic cellular morphology. Both ResNet18 and the baseline DinoBloom performed well, with the ensemble model achieving the highest F1 score and overall performance. Interestingly, the ViT failed to converge, likely due to limited data and weaker inductive biases for local structure.

We observe similar trends with H&E slides, as ResNet18 achieved perfect classification, tied with the baseline UNI foundation model, while the ViT and ensemble model did not fare as well. Overall, these results suggest that great PBS and H&E classification can be achieved by solely relying on local cellular morphologies, through architectures such as ResNet18, as it was able to stay competitive with state-of-the-art foundation models on identical training data.

Cross-Domain Classification. Models trained on PBS found significantly better accuracies on cross-domain classification than H&E-trained models, with DinoBloom achieving comparable results with its intra-domain performance across all metrics. This indicates that learned morphological information is much easier to translate from blood smears to solid tissue, despite having less available training data in the APL dataset.

Interestingly, all models trained on tissue H&E slides achieved high precision but low recall, implying that they become extremely conservative and predict the `immune` class only under high-confidence conditions. H&E-trained ViT showed the best performance across all evaluation metrics, even when compared to the PBS-trained ViT, since transformers tend to learn better transferable representations than other architectures on larger amounts of data, focusing more on global pattern and less on biases such as image texture. The performance of tissue-trained models suggest the lack of full variability of immune cell morphology when contextual cues are absent. UNI performed the worst under cross-domain evaluation, suggesting that UNI representations are inherently context-dependent, as their pretrained-weights were derived from tissue-level images without masking, where spatial organization and microenvironmental context are present.

In conclusion, these results imply that even when all tissue contexts are removed through masking, liquid biopsies retain enough variability in visual information to enable deep-learning models, such as DinoBloom, to learn transferable, morphological features and distinguish immune cells from non-immune cells. This performance, however, is contingent on training foundation models, or equivalently large networks, with massive amounts of blood smear images. Models trained on H&E tissue images exhibited limited generalization to PBS data, indicating that learned representations are strongly influenced by contextual cues. As a result, these models perform poorly on blood smear images, where isolated cells lack the rich contextual information present in tissue.

Code and Data Availability

Immunocto is accessible on Zenodo https://zenodo.org/records/11073373 and APL through Kaggle https://www.kaggle.com/datasets/eugeneshenderov/acute-promyelocytic-leukemia-apl. Relevant code is publicly available at https://github.com/hoseok-lee/immune-classifier.

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